

COER UNIVERSITY, ROORKEE

Department of Computer Science & Engineering

School of Computing

LABORATORY REPORT

Subject:

Mastery in Data Analytics and Visualizations
and Career Advancement

Experiment No. 4

Advanced Data Visualization

using Matplotlib and Seaborn

Academic Session: 2026-27

Name: UTSAV KUMAR KUSHWAHA Roll No.: 63

CUID: CU240251668 Section: B

Date of Submission: 1 SEPTEMBER

Submitted By: UTSAV KUMAR KUSHWAHA

Submitted To: AKSHAY JUNEJA

Faculty Signature: _____

Table of Contents

Table of Contents	2
1. Aim.....	4
2. Purpose.....	4
2.1 Why Visualization Matters	4
2.2 Matplotlib versus Seaborn.....	4
2.3 Dataset Context	4
3. Prerequisites	5
4. Software and Libraries Required	5
5. Dataset Used.....	5
6. Step-by-Step Procedure.....	5
Step 1–2: Load Data and Import Libraries.....	6
Step 3: Bar Charts for Categorical Comparison.....	6
Step 4: Line Charts for Temporal Trends	7
Step 5: Histograms for Distribution Analysis	7
Step 6: Box Plots for Outliers and Group Comparison.....	7
Step 7: Scatter Plots for Bivariate Relationships	8
Step 8: Correlation Heatmap.....	8
Step 9: Customization	9
Step 10: Interpretation and Insight Summary	9
7. Key Insights Derived from Visualizations.....	9
8. Chart Selection Guide	10
9. Viva Questions (Detailed Answers).....	10
Q1. Why is data visualization considered an essential component of data analytics?.....	10
Q2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?	10
Q3. Which type of chart would you choose to compare sales across different regions? Justify your answer.	10
Q4. What information can be obtained from a histogram and a box plot?	10
Q5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?.....	11
Q6. What is a correlation heatmap? How can it assist in feature selection for machine learning?	11
Q7. Why is chart customization (titles, labels, legends, colors) important in data visualization?.....	11
Q8. What factors should be considered while selecting an appropriate visualization for a dataset?.....	11
Q9. How can misleading visualizations affect business decision-making? Give a real-world example.	11

Q10. Explain how data visualization supports storytelling and business intelligence in organizations..	11
10. Conclusion	12

1. Aim

The aim of this laboratory experiment is to design, create, and interpret a comprehensive suite of data visualizations using the Python libraries Matplotlib and Seaborn. Working with a real-world business dataset (Superstore Sales or equivalent), students will produce bar charts, line charts, histograms, box plots, scatter plots, and correlation heatmaps. Each visualization will be customized with titles, axis labels, legends, annotations, and professional colour palettes. The ultimate objective is to extract and communicate clear business insights that would be difficult to obtain from tabular data alone.

Data visualization is the bridge between numerical analysis and human understanding. A well-chosen chart can reveal trends, outliers, distributions, and relationships within seconds, enabling faster and more confident decision-making. Mastery of Matplotlib (low-level control) and Seaborn (high-level statistical graphics) equips analysts with the full spectrum of tools required for both exploratory analysis and publication-quality reporting.

2. Purpose

2.1 Why Visualization Matters

Raw tables of numbers overwhelm working memory; graphical encodings exploit the human visual system's ability to detect patterns pre-attentively. Visualization therefore accelerates insight discovery, improves communication with non-technical stakeholders, and reduces the risk of overlooking critical signals such as seasonal peaks or loss-making product lines.

2.2 Matplotlib versus Seaborn

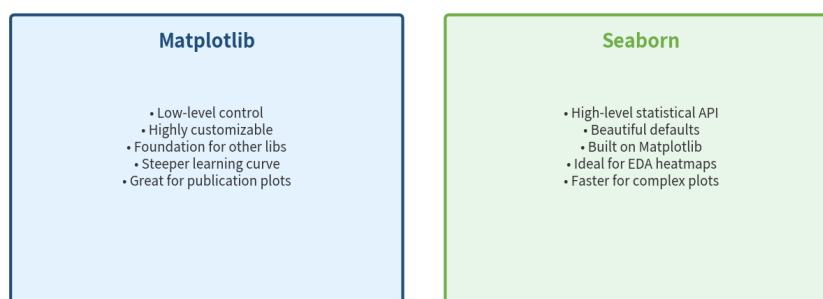


Figure 2.1 – Complementary strengths of Matplotlib and Seaborn

Matplotlib provides fine-grained control over every visual element and serves as the foundation upon which Seaborn is built. Seaborn offers a concise, statistically oriented API with attractive default styles, making it ideal for rapid exploratory heatmaps, distribution plots, and categorical comparisons. In professional practice the two libraries are used together: Seaborn for quick statistical views and Matplotlib for final polishing and custom annotations.

2.3 Dataset Context

The Superstore Sales Dataset supplies continuous measures (Sales, Profit, Quantity, Discount) and categorical dimensions (Category, Region, Segment). This mixture is ideal for demonstrating the full range of chart types covered in the experiment.

3. Prerequisites

- ✓ Basic Python programming and familiarity with Jupyter Notebook or Google Colab.
- ✓ Pandas for data loading and simple aggregation.
- ✓ Prior exposure to Matplotlib and Seaborn plotting functions.
- ✓ Completed data-preprocessing and EDA experiments with a cleaned dataset ready for visualization.

4. Software and Libraries Required

Component	Purpose
Python 3.8+	Core language for the visualization pipeline
Pandas	Data loading, filtering, and aggregation before plotting
Matplotlib	Low-level plotting engine; titles, labels, annotations, custom layouts
Seaborn	High-level statistical graphics (heatmaps, box plots, count plots)
Jupyter / Colab	Interactive notebook environment for iterative design

5. Dataset Used

Superstore Sales – Data Preview for Visualization					
Sales	Profit	Quantity	Category	Region	Discount
261.96	41.91	2	Furniture	South	0.00
731.94	219.58	3	Furniture	West	0.00
14.62	6.87	2	Office Supp.	South	0.20
907.15	90.72	5	Technology	West	0.00
22.37	-15.15	3	Office Supp.	Central	0.40

Figure 5.1 – Sample rows from the Superstore dataset used for visualization

Key fields visualized in this experiment include Sales, Profit, Quantity, Discount, Category, and Region. The combination of continuous and categorical variables enables every major chart type to be demonstrated on the same coherent business dataset.

6. Step-by-Step Procedure

Advanced Visualization Experiment Workflow



Figure 6.1 – Advanced visualization experiment workflow

Core Chart Types Used in This Experiment

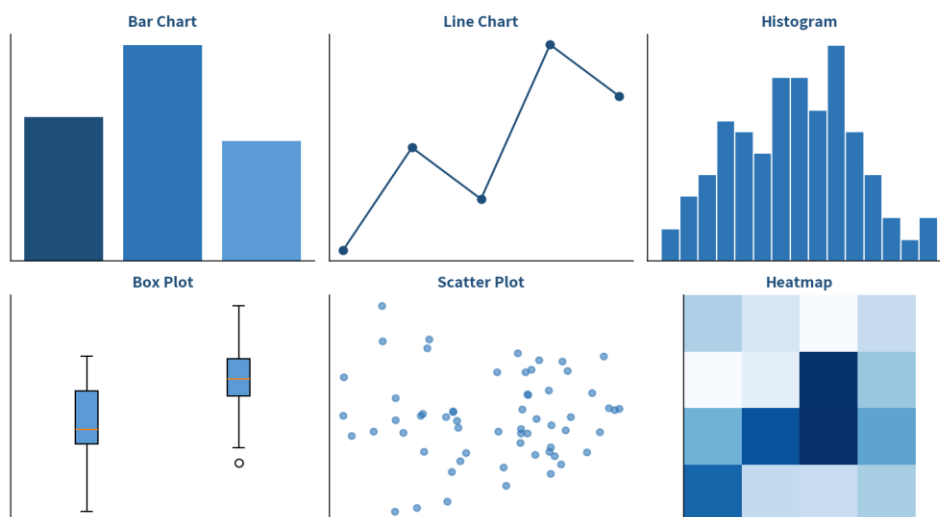


Figure 6.2 – Core chart types produced in this experiment

Step 1–2: Load Data and Import Libraries

The cleaned Superstore CSV is loaded with Pandas. Matplotlib and Seaborn are imported; a consistent style (e.g., seaborn-v0_8-whitegrid) is set for a professional appearance.

Step 3: Bar Charts for Categorical Comparison

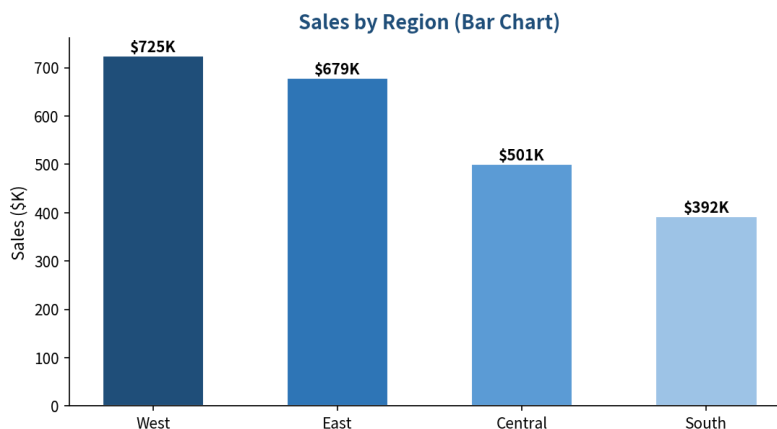


Figure 6.3 – Sales by Region (vertical bar chart)

Bar charts are the natural choice when comparing a numeric measure across discrete categories. The West region leads in sales, followed by East, Central, and South. Colour encoding and value annotations improve readability.

Step 4: Line Charts for Temporal Trends

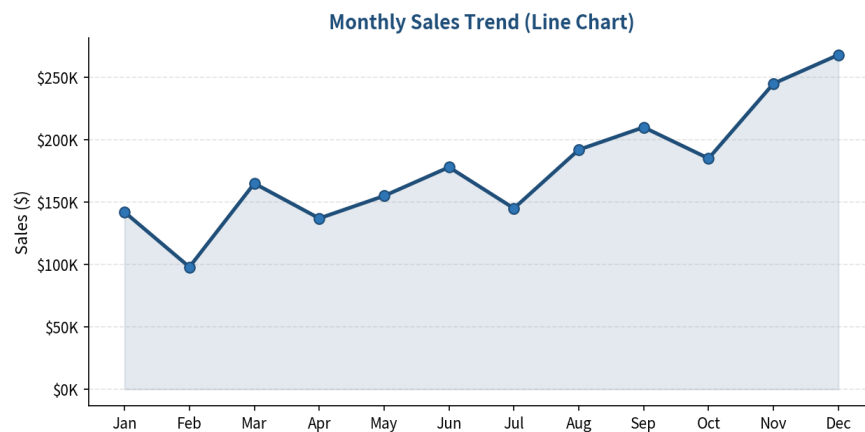


Figure 6.4 – Monthly Sales Trend (line chart with markers and fill)

Line charts reveal the evolution of a continuous measure over time. Sales exhibit a clear upward trajectory through the year, peaking in Q4—consistent with holiday purchasing patterns.

Step 5: Histograms for Distribution Analysis

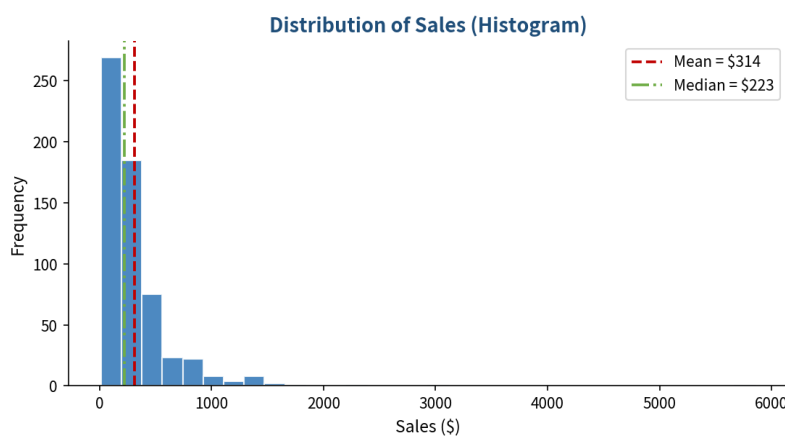


Figure 6.5 – Distribution of Sales with mean and median markers

Histograms expose the shape of a continuous variable. Sales are strongly right-skewed; the mean exceeds the median, indicating a long tail of high-value orders.

Step 6: Box Plots for Outliers and Group Comparison

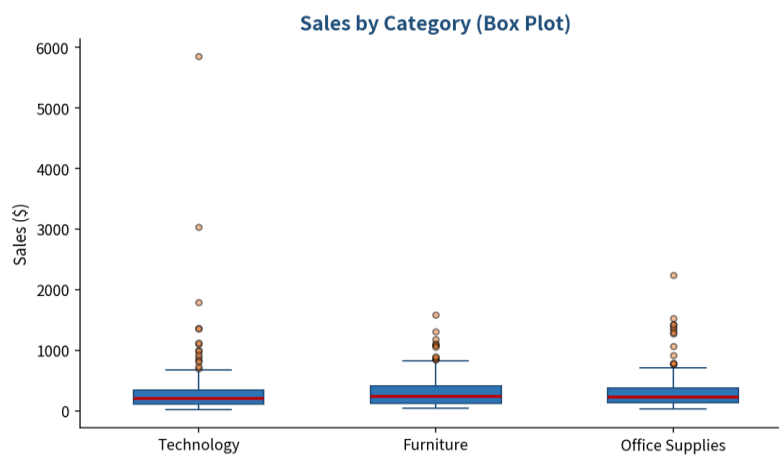


Figure 6.6 – Sales distributions by Category (box plot)

Box plots summarize the five-number summary and highlight outliers. Technology shows a higher median and a longer upper whisker, confirming its contribution of large-ticket orders.

Step 7: Scatter Plots for Bivariate Relationships

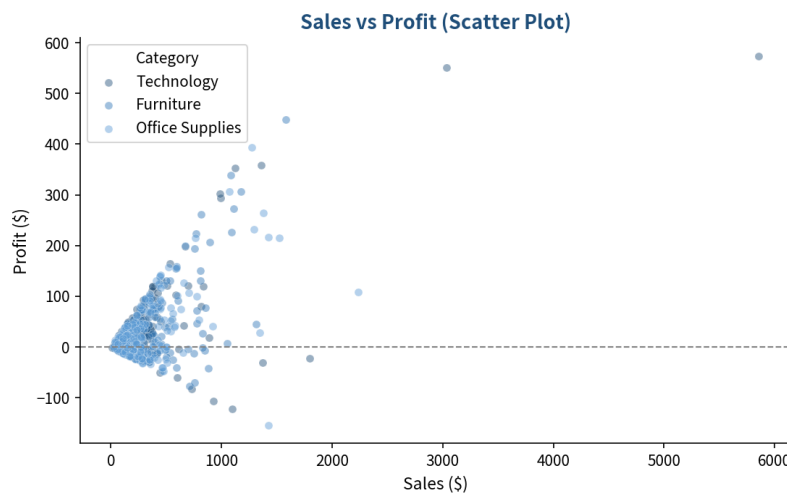


Figure 6.7 – Sales versus Profit coloured by Category

Scatter plots display the joint distribution of two continuous variables. A positive Sales–Profit relationship is visible, while points below the zero-profit line flag loss-making transactions, often associated with high discounts.

Step 8: Correlation Heatmap

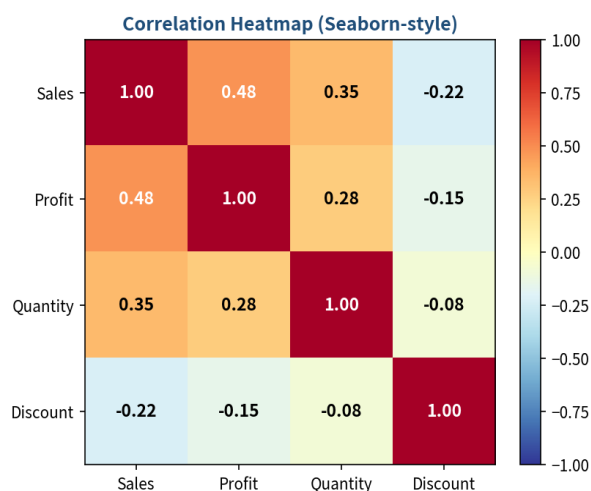


Figure 6.8 – Pearson correlation heatmap among numeric features

A Seaborn heatmap visualizes the correlation matrix at a glance. Sales and Profit are moderately positively correlated; Discount shows a negative association with Profit—an important signal for pricing policy.

Step 9: Customization

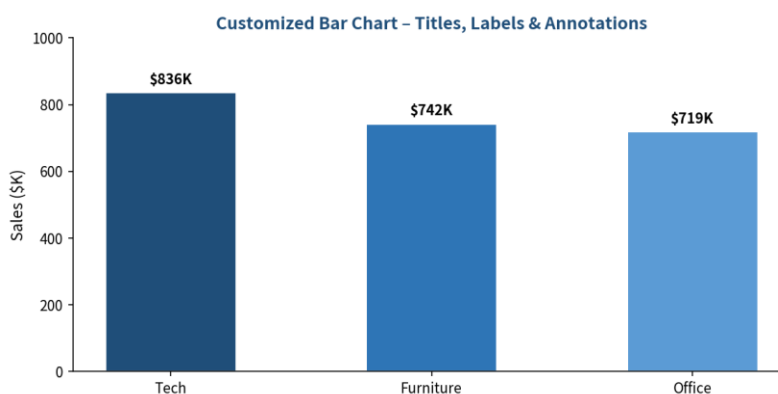


Figure 6.9 – Example of a fully customized bar chart with annotations

Professional visualizations require clear titles, axis labels, legends, consistent colour palettes, and data-point annotations. These elements transform a raw plot into a self-explanatory communication artefact.

Step 10: Interpretation and Insight Summary

Each chart is interpreted in business language. Findings are collated into a concise insight list that can be presented to stakeholders or used as input for subsequent modelling experiments.

7. Key Insights Derived from Visualizations

- West and East regions dominate sales volume; South lags and warrants further investigation.
- Monthly sales rise steadily and peak in Q4, confirming strong seasonality.
- Sales distribution is heavily right-skewed; a minority of orders drive a large share of revenue.
- Technology category contributes higher-value orders; Office Supplies dominate order count.
- Discount is negatively associated with Profit—aggressive discounting can destroy margin.

8. Chart Selection Guide

Chart Type	Best Used For	Example in This Experiment
Bar Chart	Comparing categories	Sales by Region / Category
Line Chart	Trends over time	Monthly Sales Trend
Histogram	Distribution of one numeric variable	Sales distribution
Box Plot	Outliers & group comparison	Sales by Category
Scatter Plot	Relationship between two numerics	Sales vs Profit
Heatmap	Correlation matrix	Feature correlations

9. Viva Questions (Detailed Answers)

Q1. Why is data visualization considered an essential component of data analytics?

Visualization converts abstract numbers into spatial and colour encodings that the human visual system processes rapidly. It reveals patterns, outliers, and relationships that remain hidden in tables, accelerates exploratory analysis, and enables analysts to communicate findings to non-technical decision-makers. Without visualization, analytics risks remaining an internal technical exercise rather than a driver of organizational action.

Q2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?

Matplotlib is a low-level plotting library that offers complete control over every graphical element; it is the foundation of the Python visualization ecosystem. Seaborn is a high-level interface built on Matplotlib that provides statistically oriented plot types, attractive default themes, and concise syntax for complex multi-variable graphics. Seaborn is generally more suitable for rapid statistical visualizations (heatmaps, distribution plots, categorical comparisons) because it requires fewer lines of code and applies sensible aesthetic defaults.

Q3. Which type of chart would you choose to compare sales across different regions? Justify your answer.

A bar chart (vertical or horizontal) is the most appropriate choice. Bars allow direct visual comparison of lengths, which the eye judges accurately. Each region becomes a discrete category on the categorical axis and sales becomes the continuous measure on the value axis. Adding data labels further improves precision. Alternative charts such as pie charts are less effective once more than a few categories are present.

Q4. What information can be obtained from a histogram and a box plot?

A histogram shows the frequency distribution of a continuous variable, revealing shape (skewness, modality), centre, and spread. A box plot compresses the same distribution into the five-number summary

(minimum, Q1, median, Q3, maximum) and explicitly flags outliers. Together they answer questions about typical values, variability, and the presence of extreme observations.

Q5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?

A scatter plot places each observation as a point in a two-dimensional plane defined by two continuous variables. Clusters, linear or non-linear trends, and outliers become immediately visible. Colour or size encodings can introduce a third variable. The plot is therefore the primary tool for diagnosing the form and strength of bivariate relationships before formal modelling.

Q6. What is a correlation heatmap? How can it assist in feature selection for machine learning?

A correlation heatmap is a colour-coded matrix of pairwise correlation coefficients. Strong positive or negative cells highlight linear associations; near-zero cells indicate independence. During feature selection, highly correlated predictors can be pruned to reduce multicollinearity, while variables that correlate strongly with the target become priority candidates for inclusion in a predictive model.

Q7. Why is chart customization (titles, labels, legends, colors) important in data visualization?

Customization transforms a raw plot into a self-contained communication artefact. A descriptive title states the message; axis labels clarify units and scales; legends decode colour or marker mappings; and a thoughtful palette ensures accessibility and professional appearance. Without these elements, even an accurate chart can be misinterpreted or ignored by its intended audience.

Q8. What factors should be considered while selecting an appropriate visualization for a dataset?

Key factors include: (1) the data types involved (categorical vs continuous), (2) the analytical question (comparison, distribution, relationship, composition, trend), (3) the number of variables to encode, (4) the audience's statistical literacy, and (5) the medium of delivery (screen, print, presentation). Matching chart type to these constraints maximises clarity and minimises cognitive load.

Q9. How can misleading visualizations affect business decision-making? Give a real-world example.

Truncated y-axes, inconsistent scales, cherry-picked time windows, or 3-D effects can exaggerate or hide differences. A classic example is a sales-growth chart that starts the y-axis at 90 % of the minimum value, making a modest 5 % increase appear dramatic. Executives acting on such a chart may over-invest in a product line whose true growth is unremarkable, leading to misallocated budgets and missed targets.

Q10. Explain how data visualization supports storytelling and business intelligence in organizations.

Visualization supplies the narrative arc of a data story: context (what the data represent), conflict (the problem or opportunity revealed by the pattern), and resolution (the recommended action). In business-intelligence platforms, interactive dashboards built from the same visual vocabulary allow managers to

explore data themselves, accelerating the feedback loop between insight and decision and embedding data-driven culture throughout the organization.

10. Conclusion

This experiment provided hands-on experience with the full spectrum of core data visualizations using Matplotlib and Seaborn. Students produced bar charts, line charts, histograms, box plots, scatter plots, and correlation heatmaps on a real-world retail dataset, customized each figure for clarity, and translated visual patterns into actionable business insights.

The skills acquired—chart selection, aesthetic customization, and visual interpretation—are foundational for every subsequent stage of the analytics lifecycle, from exploratory data analysis through executive storytelling. Proficiency with both low-level (Matplotlib) and high-level (Seaborn) libraries ensures analysts can move fluidly from rapid exploration to polished, publication-ready graphics.

— *End of Laboratory Report* —

COER University, Roorkee | Department of Computer Science & Engineering